Problem Set 7

1. Principal McDaniel is also monitoring population movement in the *East Wing* of the Ida B. Wells High School. Just like in the West Wing, all the hallways connecting the East Wing classrooms have been converted to be one-way traffic, and each one-way hallway has a camera detecting the number of times people walk by. The following matrix represents movement in the East Wing. When the matrix A is multiplied by a camera data vector $\vec{x}$, the result vector $\vec{\Delta}$ is the corresponding change in population vector.

$$A = \begin{bmatrix} -1 & 0 & 0 & 0 & -1 & 1 & 1 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 \end{bmatrix}$$

   a. Sketch a diagram of what the hallway system and camera system could look like based on this matrix.

   b. Find the change in room populations corresponding to the camera data vector $\vec{v} = \begin{bmatrix} 2 \\ 2 \\ 0 \\ 3 \\ 0 \\ 2 \\ 0 \end{bmatrix}$. Check your answer using both the map of the hallways and matrix multiplication.

   c. Find the set of all vectors $\vec{c}$ such that $A\vec{c} = \begin{bmatrix} 3 \\ 2 \\ 1 \\ -4 \\ -2 \end{bmatrix}$. Determine if this is a subspace of $\mathbb{R}^5$ and explain.

   d. Determine the *set U of all* possible camera data vectors $\vec{x}$ that correspond to a zero population change in each room. Also determine if U is a subspace of $\mathbb{R}^7$ or not. Explain your reasoning.

2. Choose **four rows** from the column space and null space worksheet and complete them. The rest of the rows are good practice for the exam.

3. Use the definition of a vector space to show that the set of 2x2 matrices with real-valued entries form a vector space. (You **do** need to verify all ten axioms and show any necessary work.)

4. Let $H = \{(x, y) \mid x^2 + y^2 \leq 49\}$, a circle with radius 7 centered at the origin.
    a. Is H closed under addition? Why or why not?
    b. Is H closed under scalar multiplication? Why or why not?
    c. Is H a subspace of $\mathbb{R}^2$? Why or why not?

5. Let A be an m x n matrix. Write a convincing argument for the following: $Nul(A)$ containing only the zero vector is equivalent to the columns of A being linearly independent. (Don't just quote the Invertible Matrix Theorem.)

6. Let H be the set of all vectors of the form $\begin{bmatrix} 2t \\ 0 \\ t \end{bmatrix}$. Show that H is a subspace of $\mathbb{R}^3$.

7. Let $A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$ and $\boldsymbol{w} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Determine if **w** is in Col *A*. Is **w** in Null *A*?

8. Do the following.

    a. Explain why the set of all polynomials of degree 3 **is not** a subspace of polynomials of degree *n*.
    b. Explain why the set of all polynomials of degree 3 or less **is** a subspace of polynomials of degree n.

9. (Extra Credit) Sydney, Ethan, Alex, and Shannon were enjoying the beautiful weather and studying for the exam. Sydney claimed that all subspaces of $\mathbb{R}^3$ were either points, lines, planes, or the whole of $\mathbb{R}^3$. Alex and Ethan went a bit further and claimed that any point, line, or plane in $\mathbb{R}^3$ was a subspace. Do you believe both claims? Why or why not? Explain which claim(s) you think are true and which claims you think are false and why.